Language-Aware Hard Negative Mining for Improved Vision-Language Representation Learning

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Motivation

Why does hard negative mining matter?

Problem with standard contrastive learning:

- CLIP-style training pushes matched (image, text) pairs together and unmatched pairs apart.
- Easy negatives (very different images/captions) provide little gradient signal.
- Hard negatives — pairs that are similar but not matched — are crucial for learning fine-grained distinctions.

The LLaVE insight (our starting point):

- Dynamically reweight negatives by their embedding-space similarity (hardness score).
- Removes the need for separate offline mining stages.
- BUT: hardness computed purely from the model's own embeddings has a flaw...

⚠️ *Two captions saying the same thing differently (e.g. "A dog runs" vs "A canine sprints") get treated as hard negatives — wasting model capacity.*

Our Approach

Language-Aware Hardness Score

Key Idea: inject an external linguistic prior (Sentence-BERT)

$$H(i,j) = \alpha \cdot sim_embed(i,j) + \beta \cdot sim_ling(i,j)$$

$$w(i,j) = \exp(H(i,j))$$

sim_embed

- Cosine similarity in the model's embedding space
- Scaled by logit temperature $\tau = 50$
- Updates each training step as model learns

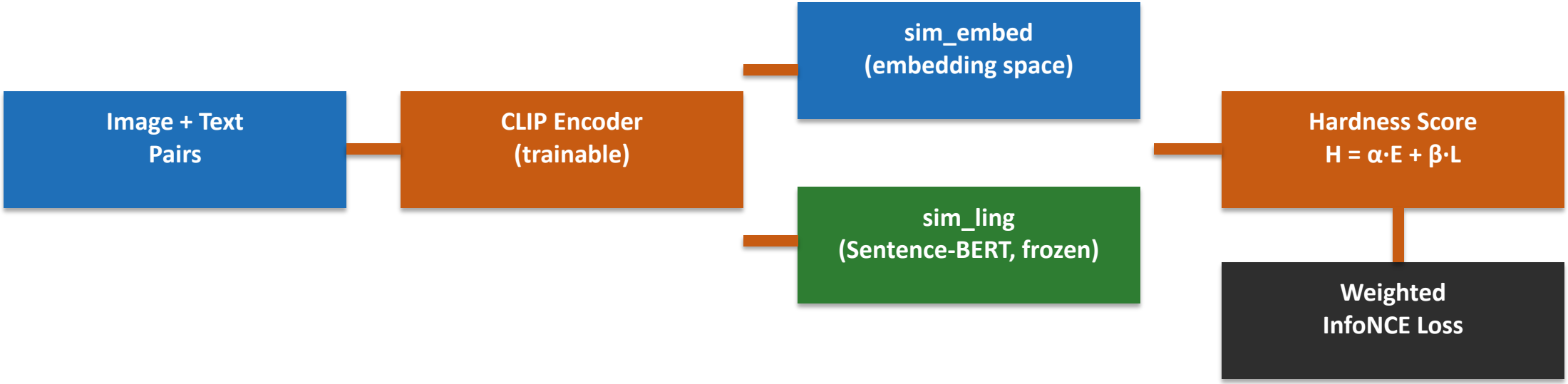
sim_ling (NEW)

- Cosine similarity from frozen Sentence-BERT
- Pre-computed once — zero overhead at train time
- Catches semantically equivalent captions

$\beta = 0 \rightarrow$ standard LLaVE baseline | $\beta = 1 \rightarrow$ our language-aware method

$\alpha = 9$ throughout all experiments

Method Overview



Pre-compute step (simularity.py):
All captions → Sentence-BERT → .pt embeddings saved to disk

At training time: batch indices → O(1) lookup of pre-computed SBERT embeddings. Zero model parameters added.

Implementation Details

Making it work in BFloat16

Two engineering fixes were needed for stable training:

Fix 1

FP32 Upcast

- BFloat16 rounds small values to 0 — loss collapses to 0/0
- Cast sim_embed and sim_ling to float32 before any exponential
- Prevents NaN gradients during training

Fix 2

Log-Sum-Exp Trick

- Naively computing exp(sim) creates huge intermediate values
- Subtract row maximum before exponentiation — numerically identical but stable
- Entire loss stays in log-space; no intermediate 1.0/1.0 rounding

Pre-computation: simularity.py encodes all captions with frozen Sentence-BERT once before training.
Result: (M × 384) tensor saved to disk. Training: batch indices → O(1) lookup. No GPU overhead.

Experimental Setup

Dataset

- Flickr30K — 1,000 test images (Karpathy split)
- 1 caption per image in evaluation
- 1,000-way retrieval: both I→T and T→I

Models

- Vision encoder: CLIP ViT-L/14 (OpenAI)
- Linguistic prior: Sentence-BERT all-MiniLM-L6-v2 (frozen)
- Logit scale $\tau = 50$

Hyperparameters

- $\alpha = 9$ (embedding similarity weight)
- $\beta \in \{0, 0.5, 1.0, 2.0\}$ (linguistic weight)
- Penalty threshold $\tau = 0.5$, strength $\gamma = 0.4$

Evaluation Metrics

- Recall@1 — correct item in top 1
- Recall@5 — correct item in top 5
- Recall@10 — correct item in top 10
- Median Rank (MedR) — lower is better
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- Reported for both directions:
- Image → Text (I→T)
- Text → Image (T→I)

Results on Flickr30K

1,000-way image-text retrieval

Method	I→T R@1	I→T R@5	I→T R@10	T→I R@1	T→I R@5	T→I R@10	Med
Baseline (β=0)	90.2	98.8	99.3	84.0	97.4	99.0	1
Ours (β=1)	**94.6**	**99.2**	**99.6**	**91.4**	**98.3**	**99.6**	**1**
Gain ↑	*+4.4%*	*+0.4%*	*+0.3%*	*+7.4%*	*+0.9%*	*+0.6%*	—

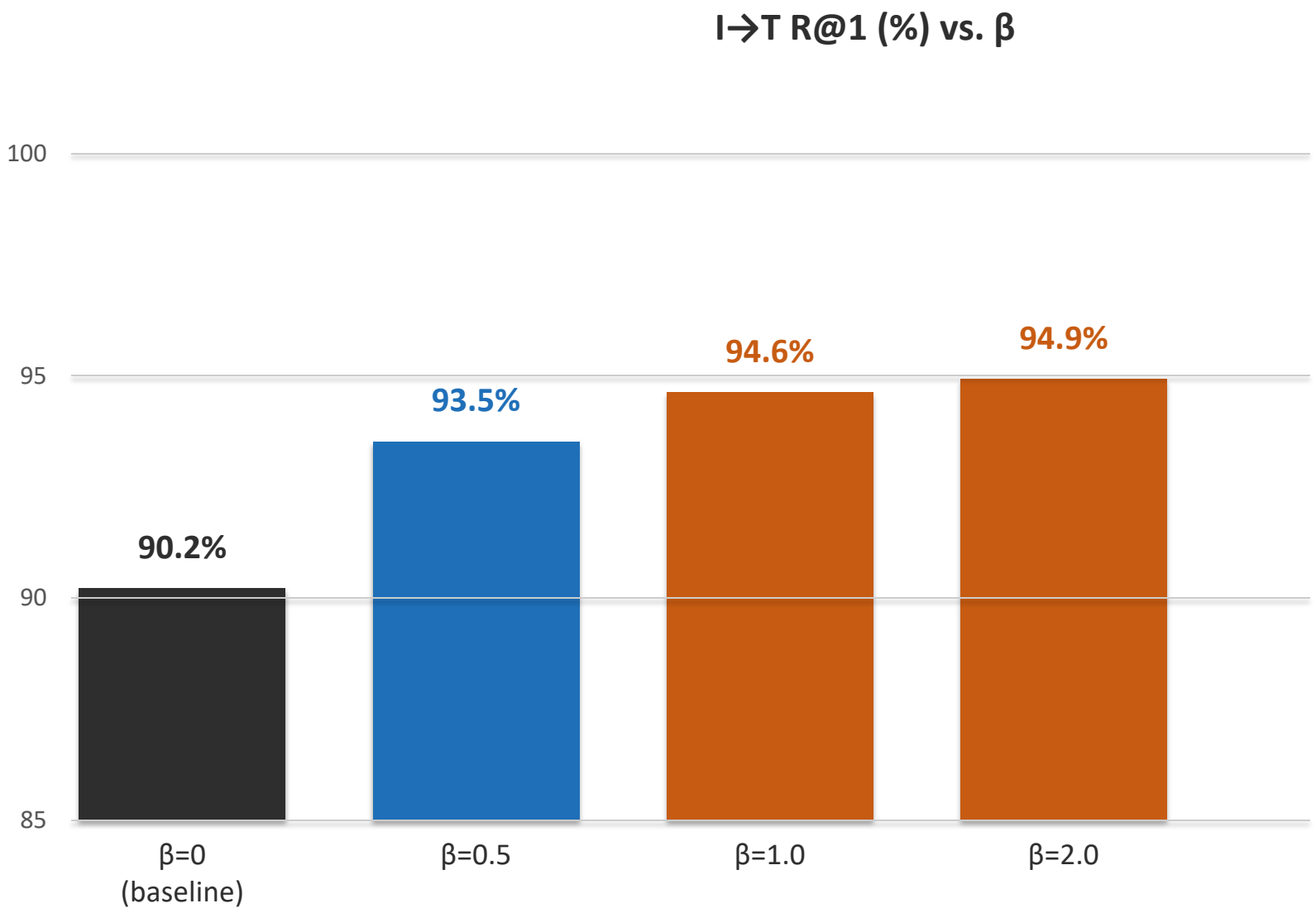
Key Takeaways:

- Largest gain at R@1: +4.4% I→T and +7.4% T→I — linguistic prior helps most at the top of the ranking.
- R@5 and R@10 already near-perfect for CLIP ViT-L/14; gains concentrated at strict top-1.
- Both MedR = 1: median correct result always appears at rank 1 in both conditions.

Interpretation: Sentence-BERT correctly identifies near-duplicate negative captions and softens their penalty — so the true match reaches rank 1 in more queries.

Ablation Study

Effect of β on I→T R@1



Observations:

- Even β=0.5 recovers most of the gain (+3.3 pp).
- Gains plateau between β=1 and β=2, showing diminishing returns.
- Method is robust: no need for careful β tuning.
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- Chosen setting: β=1 for all main results (clean round number, near-optimal).

Conclusion & Future Work

What we did:

- Extended LLaVE's hardness-weighted contrastive loss with a frozen Sentence-BERT linguistic prior.
- Combined embedding similarity and text semantic similarity into one unified hardness score.
- Added FP32 upcast + log-sum-exp for BFloat16 training stability.
- Achieved +4.4% / +7.4% R@1 gain on Flickr30K with zero extra learnable parameters.

Future directions:

- Scale to full Flickr30K + MS-COCO training sets (GPU cluster).
- Explore multimodal linguistic prior (image + caption jointly).
- Learnable α and β rather than fixed hyperparameters.
- Apply to cross-lingual retrieval where semantic overlap is harder to detect.

Core message: A frozen external text encoder can provide a complementary semantic signal that prevents semantically equivalent captions from being treated as hard negatives — improving top-1 retrieval precision with no additional training cost.